CIT2241 BUSINESS INFORMATION SYSTEMS

SEMESTER FINAL EXAM

TUESDAY, 3 JUNE 2025

09:00 – 12:00 HOURS

TIME ALLOWED: WRITING – THREE HOURS | READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is compulsory and must be attempted.
2. Section B: Answer THREE (3) questions from this section.
3. This examination paper carries a total of 100 marks.
4. Candidates must not turn this page until the invigilator tells them to do so.

Questions 1

Kafue Agro Solutions Ltd is a mid-sized agribusiness company in Zambia that supplies seeds, fertilizers, pesticides, and farming equipment to farmers across both rural and urban areas. Since its establishment in 2015, the company has experienced significant growth and now operates through a network of field agents, regional retail outlets, and a basic mobile ordering system. Its customer base includes individual small-scale farmers, farming cooperatives, and agricultural training centers.

However, the company continues to rely on manual and paper-based systems for managing key operations such as inventory tracking, sales reporting, and customer data collection. Field agents submit handwritten reports that must be manually compiled at the head office, leading to delays and errors. These outdated systems have resulted in frequent stockouts, inaccurate sales data, and poor visibility into customer purchasing trends. Consequently, this has led to customer dissatisfaction, missed sales opportunities, and slow decision-making at management level.

Kafue Agro's departments operate largely in isolation, with little integration between sales, inventory, logistics, and finance. For example, the procurement team often overorders or underorders due to a lack of real-time stock data, while the finance team struggles to produce timely reports because of fragmented records.

Recognizing these inefficiencies, the company's leadership is considering adopting Business Information Systems (BIS) to streamline operations, improve data accuracy, and enhance customer service. Their goals are to achieve real-time inventory management, strengthen internal communication, provide faster customer support, and enable more informed business decisions.

At the same time, the leadership is concerned about data privacy, cyber threats, and the cost and complexity of implementing new systems. They are particularly interested in affordable, scalable, and user-friendly solutions that can be adopted gradually across the company. Training employees, especially field agents with low digital literacy, is also an important consideration.

a) Identify and explain five types of Business Information Systems that Kafue Agro Solutions Ltd can implement to improve inventory and sales tracking.

(10 Marks)

b) Discuss how a Management Information System (MIS) could support better decision making at Kafue Agro Solutions Ltd.

(10 Marks)

c) Describe five data security measures the company should implement to protect customer and transaction data on its digital platform.

(10 Marks)

d) Identify five ethical issues that might arise from collecting and using customer data and suggest how the company can address them.

(10 Marks)

[Total: 40 Marks]

SECTION B: Attempt any THREE questions in this section

Question 2

a) With the increasing use of digital platforms to collect and store personal information, data privacy has become a critical concern. Discuss the importance of data privacy in modern information systems and explain two measures organizations can take to protect users' personal data. **(7 Marks)**

b) The Freedom of Information Act is a law that gives citizens the right to access information held by public authorities, promoting transparency and accountability in governance. Explain the purpose of the Freedom of Information Act and discuss **two ways** in which it empowers citizens to participate in democratic governance. **(7 Marks)**

c) Identity theft occurs when someone unlawfully obtains and uses another person's personal information, often for financial gain or to commit fraud. Suggest preventive measures individuals can take to protect themselves. **(6 Marks)**

[Total: 20 Marks]

Question 3

a) You are a sales manager tracking the monthly sales (in units) made by five sales representatives. The data below shows the number of units sold by each representative in January.

Sales Representative	Units Sold
Alice	120
Brian	150
Clara	135
David	110
Emily	145

Task:

1. Draw a spreadsheet layout in your answer script showing the data above using appropriate column headers.
2. Write the Excel formulas to:
    - **Calculate the total number of units sold** using the SUM function.
    - **Calculate the average number of units sold** using the AVERAGE function.

Instructions:

- Clearly label your columns and rows.

- Assume the data is entered in **cells A2 to B6**.
- Show the formulas that would be entered in the cell below the data (e.g., in **B7** and **B8**)

(10 Marks)

b) The Zambian government has adopted e-Government initiatives to improve service delivery, transparency, and citizen engagement by leveraging digital platforms. Identify and describe two **key e-Government services** currently provided in Zambia. For each service, explain how it enhances efficiency and accessibility for citizens or businesses. **(10 Marks)**

[Total: 20 Marks]

Question 4

a) An employee at a financial services firm notices that sensitive emails containing client data are being read by an unauthorized third party. However, no changes have been made to the contents of the emails, and the communication flow remains uninterrupted. Later, the firm discovers that a hacker has also altered some transaction logs in the database to hide fraudulent activity.

Based on the scenario above, identify which part of the attack is passive and which is active. Justify your classification by explaining the characteristics of each attack type as demonstrated in the scenario. **(8 Marks)**

b) A medium-sized company recently suffered a data loss incident after a power outage, and unauthorized individuals were found loitering in restricted areas of the building. Upon investigation, it was discovered that there was no backup system in place, the UPS had failed, and access to sensitive areas was loosely monitored with no sign-in or surveillance systems.

Based on the scenario above, analyze the weaknesses in both physical security and access control measures, and propose at least two specific controls. **(8 Marks)**

c) A Transaction Processing System (TPS) is a type of information system designed to collect, store, process, and retrieve large volumes of data from routine transactions in an organization. List any four types of transaction processing systems. **(4 Marks)**

[Total: 20 Marks]

Question 5

a) Information quality refers to how good, accurate, and useful information is for its intended purpose. In simple terms, it measures how trustworthy and effective the information is when people use it to make decisions or complete tasks. Identify and explain any three characteristics of information quality that can improve decision making. **(7 Marks)**

b) Business Intelligence (BI) tools are widely used to support data-driven decision-making across industries. Evaluate how the use of BI tools influences strategic decisions in organizations, and justify your answer with relevant industry examples. **(6 Marks)**

c) In high-stakes decision-making, the quality and completeness of data play a critical role in shaping outcomes and minimizing risk. Evaluate the role of data completeness in high-stakes decision-making and explain how missing information can influence strategic outcomes. **(7 Marks)**

[Total: 20 Marks]

END OF EXAMINATION